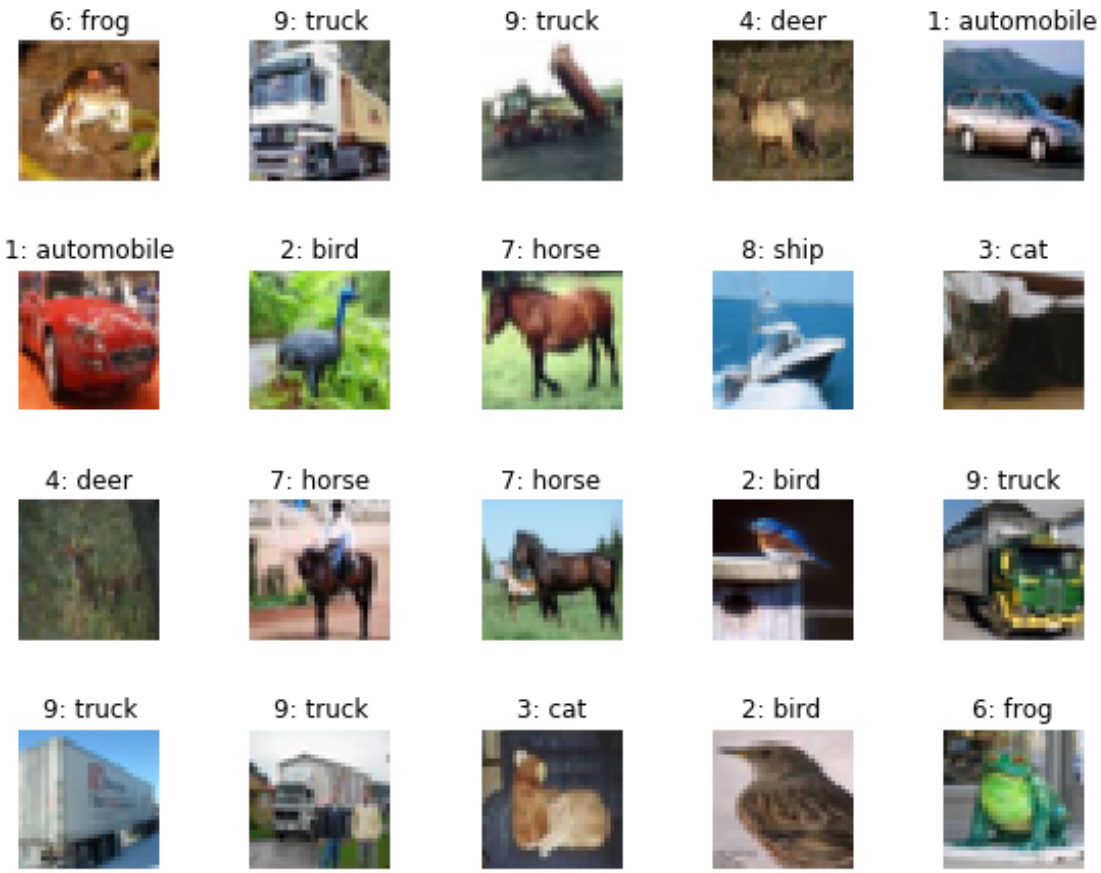
Convolutional neural networks

Ens'IA

Ensimag 2021-2022

15 novembre 2021

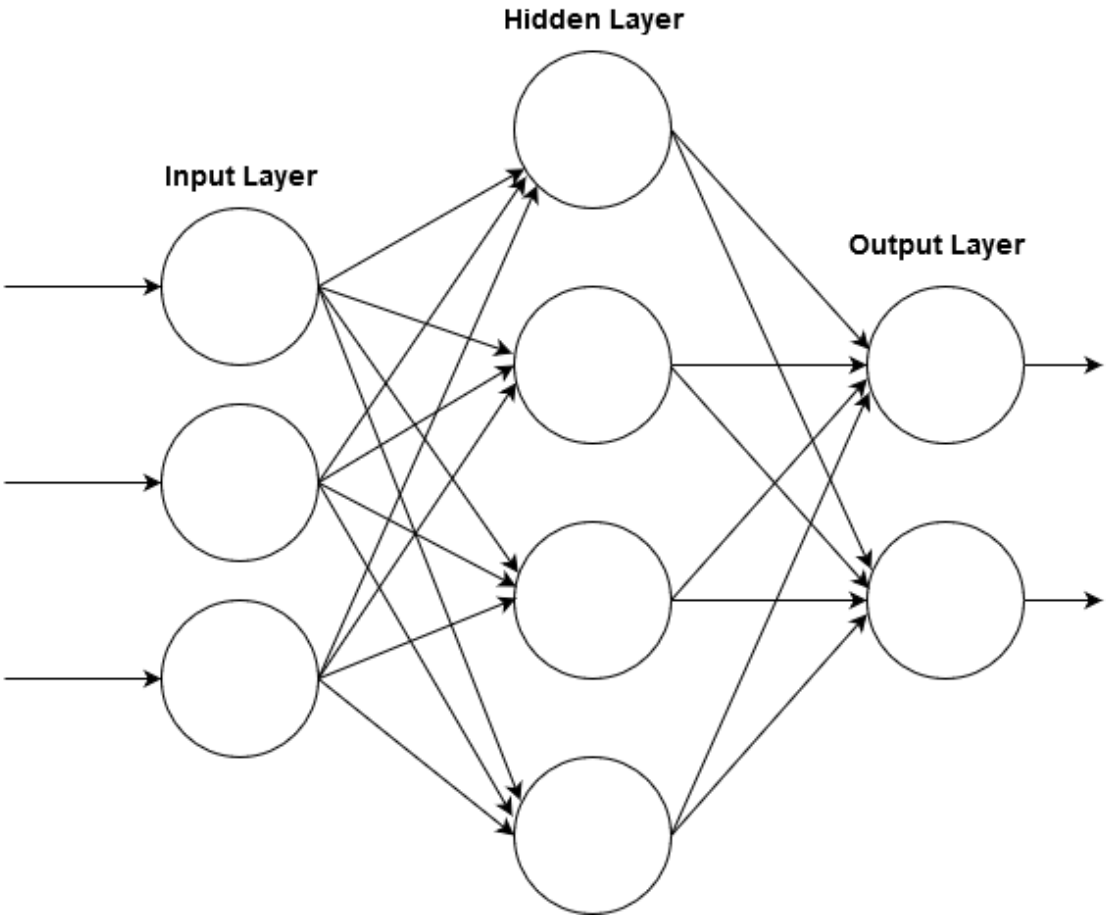
Reminder



Images classification
$\rightarrow$ We need a function f such that $f(x) = y$

How to approximate this function ? $\rightarrow$ Neural networks

Neural network



→ Succession of neuron layers

Convolution

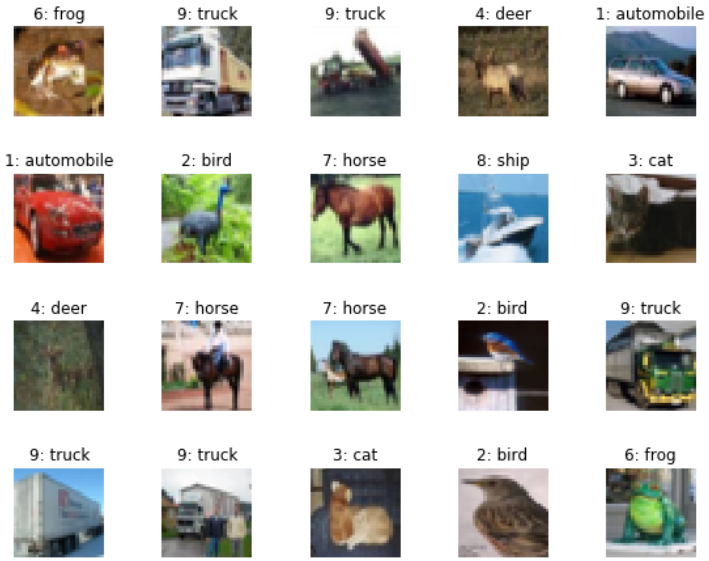
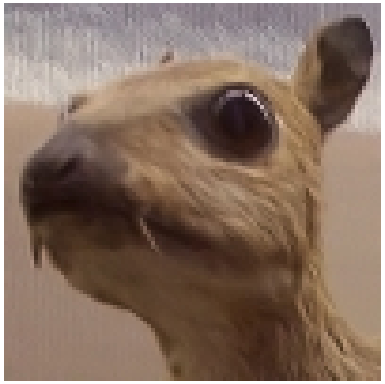


FIGURE 1 – CIFAR-10 vs. ImageNet

Convolution

Idea :

→ Image processing technique : find features in the image



Convolution

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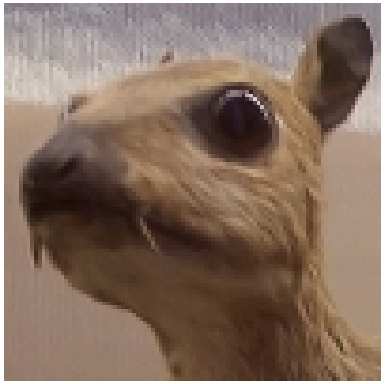
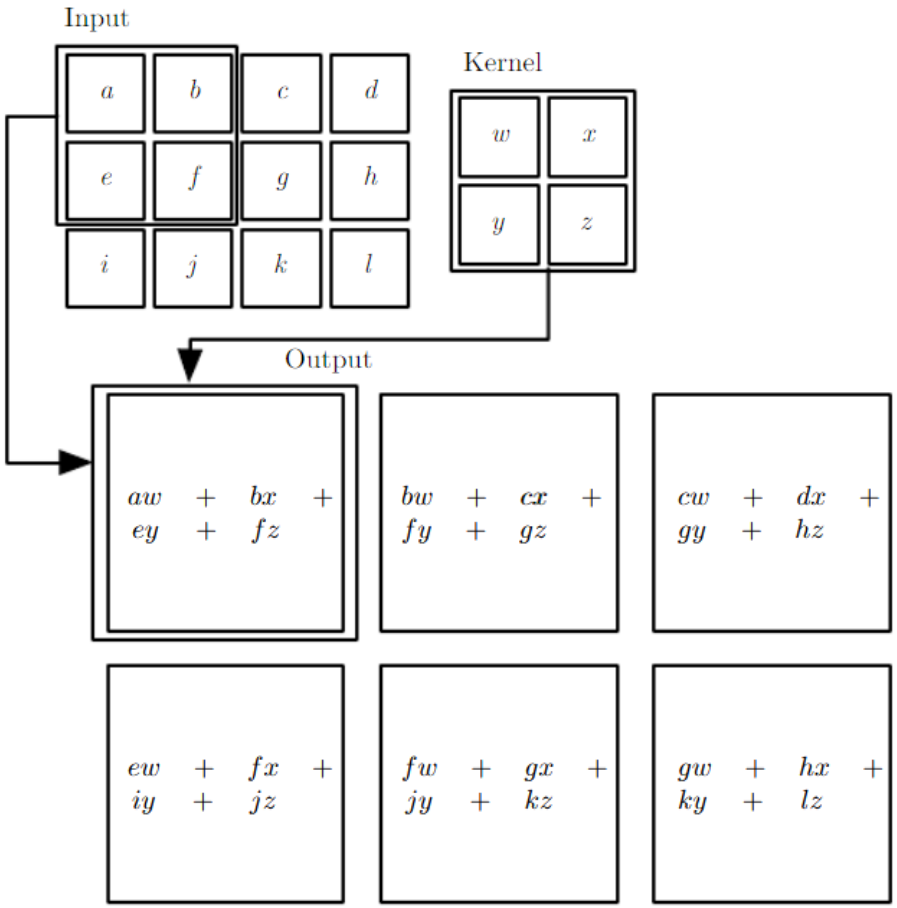


How to do that ?
→ Apply filters : a convolution kernel

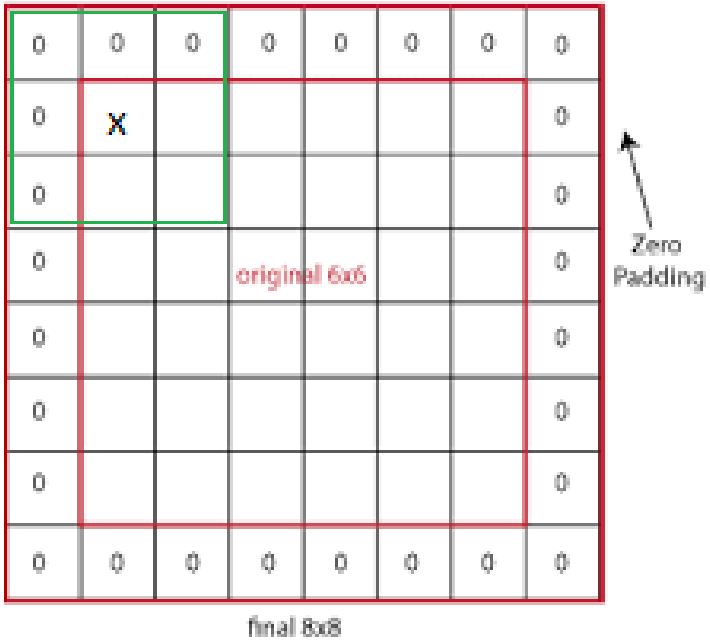
$$H = \begin{pmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{pmatrix}$$

Convolution

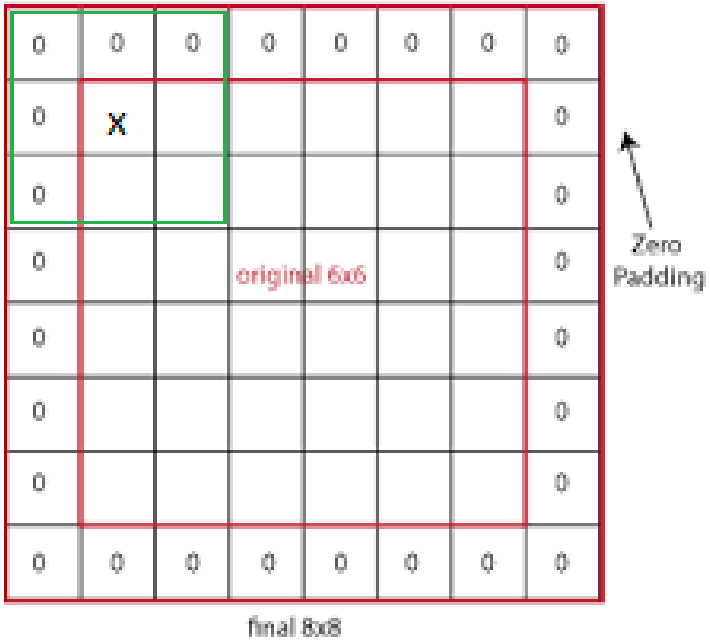
→ The mathematical operation



Convolution



Convolution



4 parameters to tune (+ the activation function) :

- The size of the filter : *kernel size*
- The number of filter : *depth*
- The number of pixels that the filter moves on each iteration : *stride*
- *zero padding*

Pooling

$\rightarrow$ Problem : dimensions can increase rapidly !

Pooling

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→ Idea : reduce the dimensions of data by combining the outputs of the previous layer

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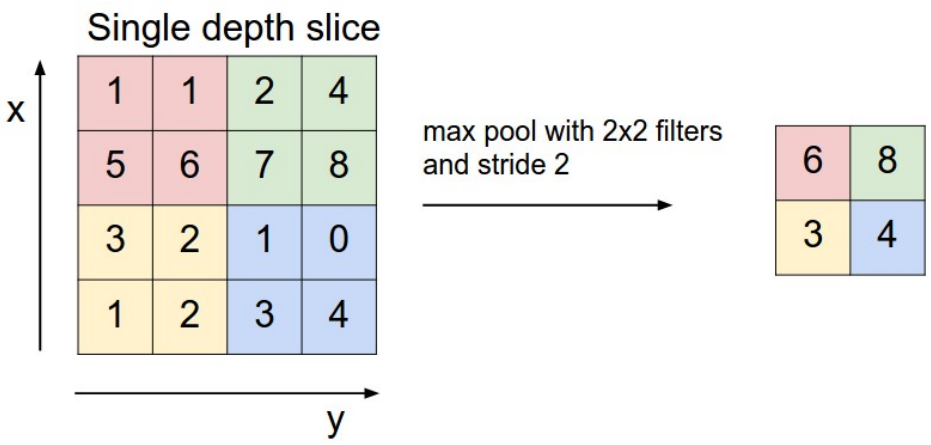
→ different types :

- *Max Pooling*
- *Average Pooling*



CNN

$\rightarrow$ After the convolutional and pooling layers, we add fully connected layers.

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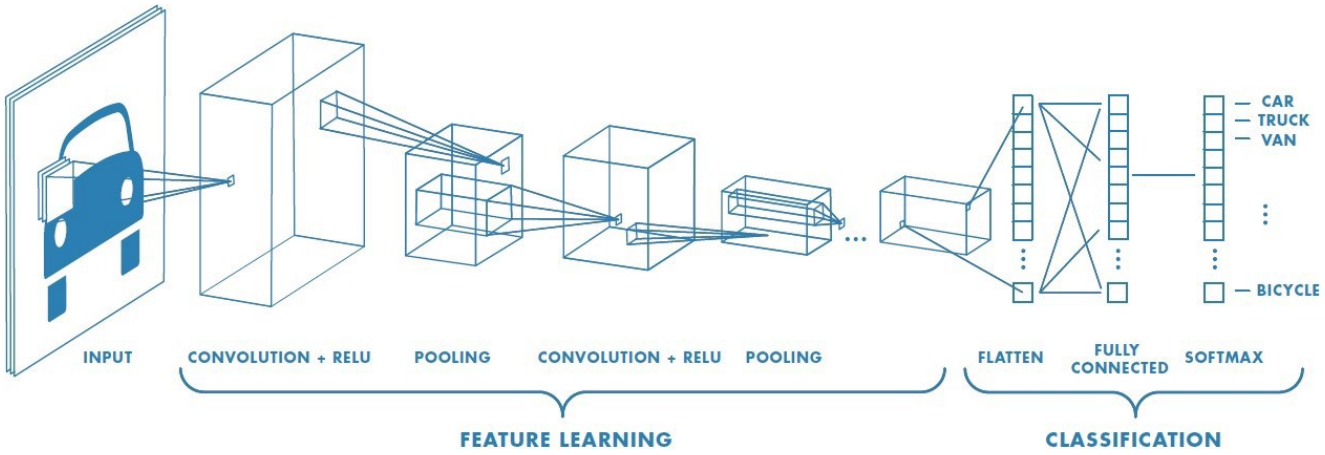
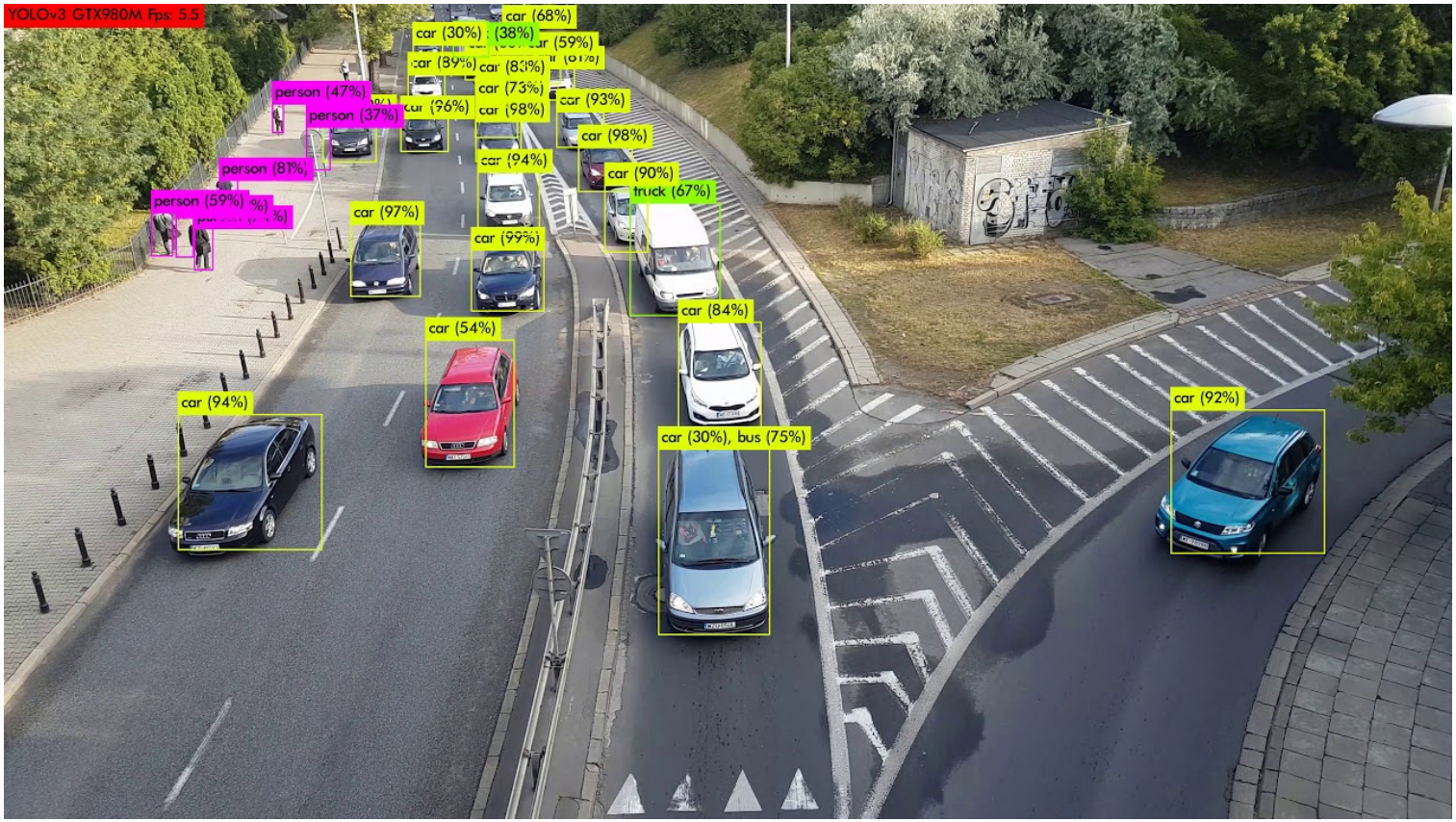
→ Summary :

- Convolution then Pooling Layers
- Fully connected layer



State of the art

Yolov3 for object detection



State of the art

Yolo =

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Yolo = You Only Look Once

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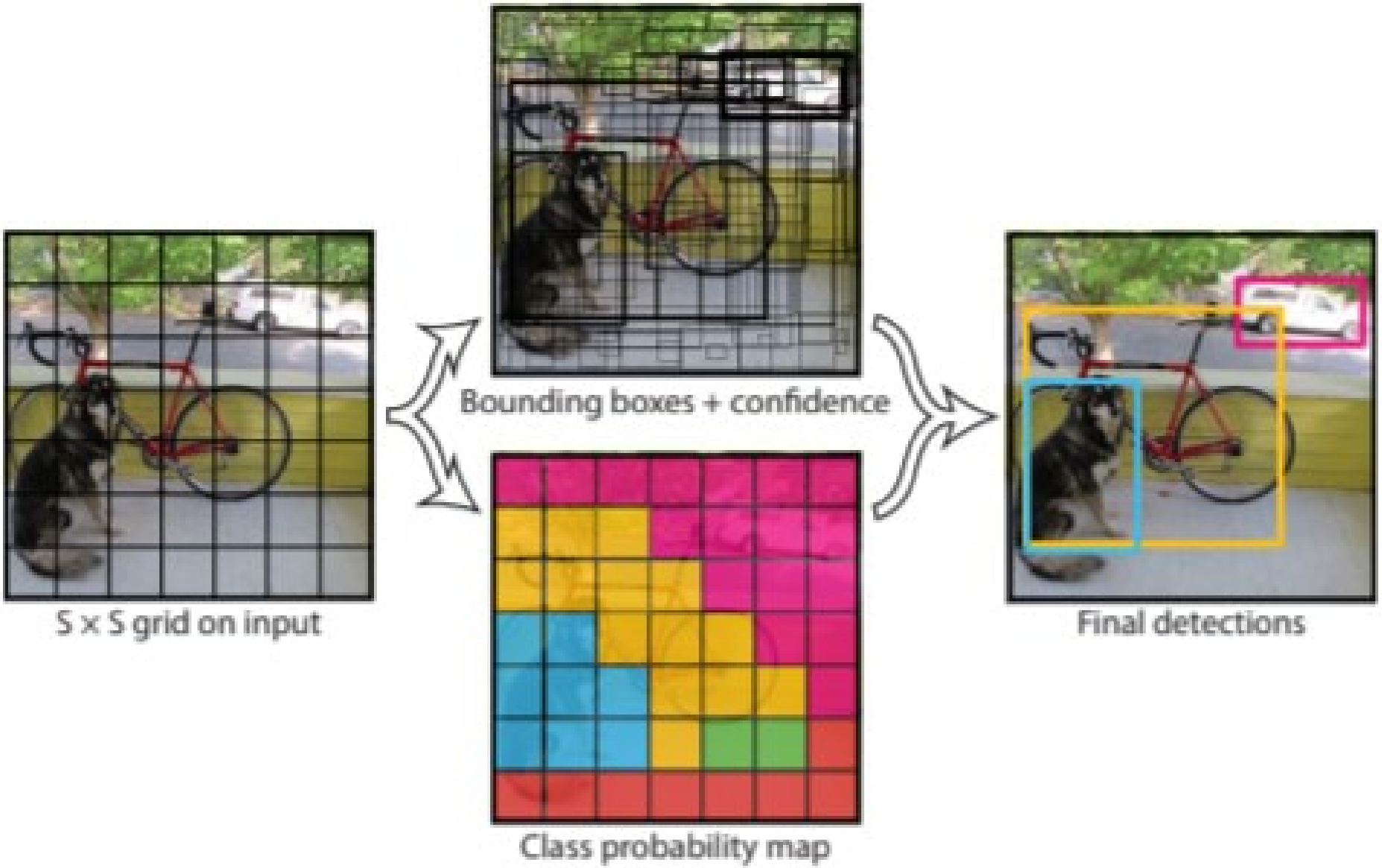
1. Residual blocks : the image is divided into various grids. Each grid has a dimension of S x S.
2. Bounding box regression : outline that highlights an object in an image.
3. Intersection over union (IOU) : a phenomenon in object detection that describes how boxes overlap.

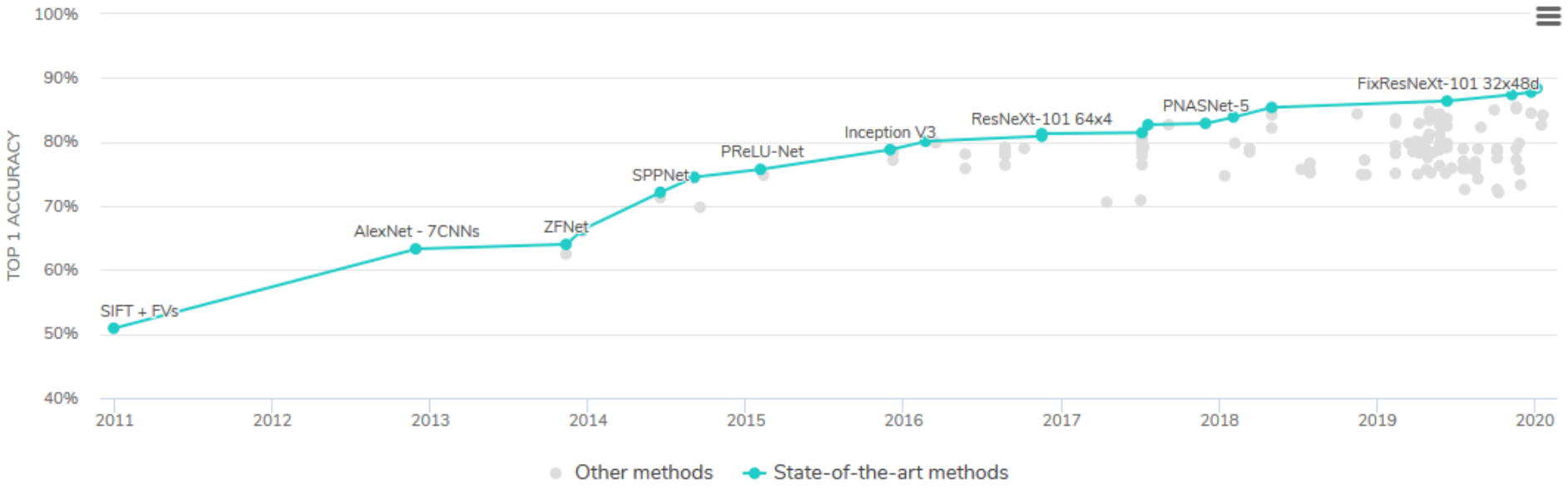
→ Only a single forward propagation through a neural network to detect objects (fast !)

State of the art



State of the art

Image Classification using ImageNet



Famous networks : ResNet, DenseNet, VGG
→ Traning for many weeks
→ very effective

Transfer Learning

Idea : Some models were trained for weeks to learn a job

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Can we "transfer" what it learned ?
$\rightarrow$ Yes.

Transfer Learning

1. Choose a model that was used for a similar job
2. Replace the fully connected layers
3. Train the fully connected layers
4. Use your new model

→ Faster
→ More effective

Sources

- CS231N (http ://cs231n.github.io/convolutional-networks/)
- http ://neuralnetworksanddeeplearning.com/
- http ://www.deeplearningbook.org/
- http ://towardsdatascience.com/
- Wikipedia
- https ://machinelearningmastery.com/transfer-learning-for-deep-learning/
- https ://paperswithcode.com/sota/
- https ://www.section.io/engineering-education/introduction-to-yolo-algorithm-for-object-detection/